

Weekly Progress Report – Week 02

Student Information

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 - **Department:** B.Tech
 - **Project Title:** Civic Website / Geo-Location Based Systems Development
 - **Reporting Week:** May 2026 – Week 02
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1. Overview

During this week, the project development focused on enhancing advanced system functionalities, improving communication flow between users and authorities, strengthening real-time updates, and optimizing overall system performance.

Major efforts were dedicated to implementing complaint monitoring mechanisms, improving dashboard usability, enhancing backend efficiency, and ensuring smoother interaction between different user roles within the application.

This phase contributed toward making the system more reliable, scalable, and user-friendly while preparing the platform for large-scale usage.

2. Tasks Completed

Dashboard Enhancement

- Improved dashboard layout for better visualization of complaint statistics.
- Added summary cards to display total complaints, pending requests, completed tasks, and active users.
- Implemented dynamic filtering options for quick data access.
- Enhanced navigation experience for different user roles.

Real-Time Complaint Tracking

- Developed real-time complaint status monitoring functionality.
- Improved live updates for complaint progress.
- Enabled users to track submitted complaints without manual refresh.
- Added timestamp-based status updates.

Officer-Worker Communication Module

- Enhanced communication flow between officers and workers.
- Implemented task assignment mechanisms.
- Added internal status comments for complaint handling.
- Improved notification handling for assignment updates.

Backend Optimization

- Improved database query performance.
- Reduced unnecessary API calls through optimized request handling.
- Refactored backend service layers for better maintainability.
- Enhanced server-side validation.

Testing and Debugging

- Conducted integration testing across modules.
 - Performed UI and functional testing.
 - Fixed bugs related to complaint updates and response synchronization.
 - Improved exception handling mechanisms.
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3. Milestones Achieved

- Successfully implemented dashboard analytics improvements.
 - Complaint tracking system became more responsive.
 - Officer-to-worker workflow was improved and stabilized.
 - Backend performance optimization reduced unnecessary processing.
 - Real-time update mechanisms showed better user experience.
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4. Challenges and Hurdles Faced

Real-Time Data Synchronization Issues

Frequent updates caused inconsistency between frontend display and backend records.

Solution Implemented

- Optimized API polling intervals.
 - Improved state management.
 - Added response synchronization handling.
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Database Performance Bottlenecks

Large complaint data affected loading performance.

Solution Implemented

- Optimized query execution.
 - Reduced redundant database calls.
 - Improved indexing strategy for faster retrieval.
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Role-Based Access Management Challenges

Different user roles required secure access restrictions.

Solution Implemented

- Strengthened route protection.
 - Improved authorization validation.
 - Added role-specific rendering conditions.
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5. Lessons Learned

This week introduced several practical and technical learning experiences.

Technical Skills Improved

- Better understanding of dashboard development.
- Improved knowledge of API optimization.
- Enhanced experience with role-based authorization.
- Learned techniques for real-time data synchronization.

Problem-Solving Skills

- Improved debugging efficiency.
- Learned methods to reduce application latency.
- Strengthened system optimization approaches.

Project Management Insights

- Understood the importance of scalable architecture.
- Learned efficient coordination between frontend and backend teams.
- Improved documentation and testing practices.

6. Future Plan

The following activities are planned for the upcoming week:

- Implement advanced analytics and reporting.
- Improve complaint priority handling.
- Add notification management features.
- Strengthen security validation mechanisms.
- Perform end-to-end testing and deployment preparation.

7. Conclusion

Overall, Week 02 was productive and focused on improving system efficiency, communication, and performance optimization. Significant progress was made toward creating a more scalable and responsive Civic Website platform. The development activities completed during this week strengthened technical knowledge, enhanced practical implementation skills, and prepared the project for future expansion.